

Sentiment Analysis for Restaurant Rating

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Abstract— The rating systems are important to find the quality of the product or service. These rating systems serve as a guide in finding the perfect one based on different user criteria. Therefore, a sentiment analysis system has been designed for automatic restaurant rating which will be useful to the people in picking their favorite restaurant. The sentiment analysis for restaurant rating system rates the restaurant depending upon the reviews given by the users. The system breaks user comments to check for sentiment keywords. Once the keywords are found, it associates the comment with a sentiment rank. Sentiment analysis can also be extended further to improve the business process. With the customers' reviews, one can understand the changes in the market and improve their product/service. It is also scalable to any type of environment.

Index Terms— Sentiment analysis, Text classification, Opinion mining, Restaurant rating, Review, Feedback.

1. INTRODUCTION

A common use case for sentiment analysis is to discover how people feel about a particular topic. This is the kind of insight one aims to find through market research which requires enormous budgets and countless man-hours to conduct surveys and cold calling. The sentiment analysis is used to get an understanding of the emotional tone behind the person's writing [1]. It finds whether the attitude of the person is positive, negative or neutral based on their writing. Opinion mining helps us to get an overall understanding of the opinion of others for certain topics like politics, books, movies, food etc. It serves as a better tool for social media monitoring.

Over the period of time, people started using Google to look for the rating of a product or a service. But the results might not be convincing. Google has been serving as an advertising platform for many products and business. The owners of these products, in order to promote them might not have provided the true information about the product and the service. This imposes a need for an automated system for product rating [2].

The rating system should be by the customers and for the customers. So the rating system is based on the customer's reviews for a restaurant [3]. Each review will be split and searched for sentiment keywords. The reviews will be classified into positive and negative and finally an overall rating will be provided to the restaurant.

The sentiment words in the reviews are checked and the highlighting given to each emotion of the customer is taken into account. The sentiment score of the review is emphasized with extra points for every highlighted emotion with emoticons, adverbs, special case idioms etc. The system is fully automated for user convenience.

2. LITERATURE REVIEW

Some of the articles by eminent scholars on product and service rating are studied and discussed below.

Gan et.al designed the system for rating a restaurant based on sentiment analysis and the between-restaurant and within-restaurant variances are explained [4]. In addition to food, service and ambience, the overall rating of the restaurant also focusses on the pricing and special contexts. The pricing refers to whether the customer is satisfied with the services provided by the restaurant for the price they are paying. In some cases, the reviews may be based on how the restaurant provides services for special occasions like birthday, anniversary etc. Here, the reviews from the casual languages are pre-processed [5]. There is no further emphasizing done on the scores based on highlighting of words in their views as they are pre-processed.

Xia et.al developed a model that extends the Dual Sentiment Analysis framework from the two class classification to three class classification, by taking the neutral reviews into consideration [6]. Reversing the sentiment context of the text is called the polarity shift. The negation being the important part of polarity shift should be handled well. This model presents the polarity shift knot in the sentiment analysis. With the sentiment reversed reviews, a data set is

formed. For every original review, a reversed review is constructed. Then a corpus-based method is developed for constructing a pseudo- antonym dictionary to reduce the dependency of dual sentiment analysis on an external antonym dictionary [6].

The content based rating system provides a numerical value within a specific range that explains the user's true sentiment towards the service/product [7]. The most important point to note in this method is that the users express both positive and negative features of the service/product in the same sentence [8]. In the service or product review, the rating annotations might be inconsistent. The misleading ratings might bring ambiguities and uncertainties in the problem. To solve these issues, Zhang et.al proposed a collaborative rating inference (CRI) model [7] which finds the correlations between the different ratings based on various aspects of user opinion for a collaborative rating inference model [9], [10].

The starting is often first seen before the textual portion of the consumer review is seen and read. Given its importance as the first-seen item in a consumer review, it is important that the star rating correctly reflects the consumer's sentiment as reflected in the review comment. Prithvirajan et.al came up with an efficient approach for exploring how the star ratings provided by the system coincides or deviates with the sentiment extracted from the text portions of the review [11].

Lak et.al developed a system to compare sentiment analysis results with star ratings in three different domains [12]. The objective of this method is to depend on star ratings when such ratings are available and as a substitute to get the rating in the cases where it is not available. The sentiment analysis not only helps us to decide which reviews to be read but also which parts of the review is to be read to get the required crisp information. Lexalytic is used as the sentiment analysis too in this system [13]. Lexalytic is a tool for systems that perform sentiment analysis or opinion mining. Lexalytic consists of a large dictionary of phrases that has sentiment values in different languages and has a sentiment score for each phrase for the analysis [14].

Anastasia et.al states that the companies under consideration, GOJEK and Grab use the social networking site twitter to reach their customers and promote their service/product based on the popularity and view of the product in the social media [15], [16]. About 126,405 tweets were collected from February to March 2016 containing GOJEK and Grab keywords. Then, the tweets regarding a particular topic were pre-processed and labelled manually and they were classified using three algorithms: Support Vector Machine, Naïve Bayes, and Decision Tree. Machine

learning classifiers were used in this system for sentiment analysis. Classifiers achieve better performance in larger dataset. Different algorithms works best in different kind of dataset. For example, Naïve Bayes works well in small dataset while Support Vector Machine performs best in large dataset. The performance of the Decision Tree algorithm is independent of the size of dataset [17].

The automatic generation of rating based on comments has also been done using the machine learning classifiers [18]. Each classifier differs in the accuracy score. The data is tested and trained with many machine learning algorithms and the corresponding accuracy scores are compared to find the best algorithm for the application [19], [20].

From the above literary survey, a system is proposed that gets the reviews from the users and provides an automated rating system that could be used as a restaurant guide and a tool to identify the changing dynamics in the market place [21]. This system is scalable to any type of environment which includes books, politics, product, etc.

This system can take a number of reviews as input for a particular restaurant and gives an extracting. Since the system focuses on the exact emotions of the customers and emphasizes the scores based on the highlighted reviews, the rating is pure. No grammar or spelling check is done for the reviews since the emoticons and expressive comments like "awww", "too GREATTT", "nice :)!!!" are taken into account for additional sentiment scores. When a new review is added, the rating is changed so that one review does not affect the entire rating of the system.

This system is low cost to maintain and an effective way to improve the business. This serves as a guide for the users to find the best product in the customer perspective.

3. SYSTEM FRAMEWORK

A. Data Collection

The proposed system is tested based on the data extracted from the Yelp Dataset. The Yelp dataset contains more than 250,000 review pictures and text information about various organizations.

To increase the trust of the information used for training, the data considered has 100,324 review texts on 2000 restaurants in several categories such as Cafe, Barbecue, Chinese, fast food as in Table 1 [22]. Each review has about 80 English words such as the, good, bad, a, like, similar etc. as shown in Table 2. The data set consists of reviews with exclamatory points, question marks, boosted words, capitalized words etc. which provides emphasis. Hence a

minimum of 50 percent of the data set consists of reviews with emphasis on it. The restaurant and its comments are scraped from the yelp dataset and the testing of the proposed system is performed on this data[23].

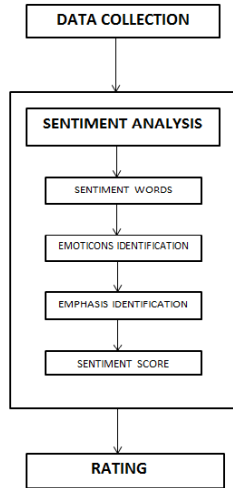


Fig. 1. System Framework

Table 1. Restaurant Category

<i>Restaurant</i>	<i>Category</i>
Ciclo Café	Cafe
BBQ	Barbecue
Cascade	Chinese
KFC	Fast food

Table 2. Sample user comments

<i>SNO</i>	<i>Comments</i>
1	Ambience is awesome and the food is really awesome. The menu has a lot of variety of good food and the service is also very good. They treat the customer politely and the place is really awesome
2	The restaurant ambience was a bit okay. The pasta tasted awful but the fish was really good. Service and ambience was okay. On the whole the restaurant was bad except for sea food which was good.

Higher score than the word ‘good’ and similarly the word ‘negative’ has a lower score than the word ‘bad’.

Table 3. Sentiment Scores for words

<i>Term</i>	<i>Score</i>
Positive terms	
Great	3.1
Good	1.9
Negative terms	
Bad	-2.5
Negative	-2.7

1. Emoticons Identification

Once the comment is broken into separate words, the punctuations are separated from the words in order to identify emoticons like :-), ;), :-(, =|, :-0, :-|, etc. Just as similar to the scores for the sentiment words, the emoticons will have their score depending on whether it adds positivity or negativity to the given comment as shown in Table 4. For example, a happy emoticon will have a higher score than asad emoticon. Now the identified emoticons will be associated with its respective score.

Table 4. Emoticons Scores

<i>Emoticons</i>	<i>Score</i>
Positive	
:-)	0.3
;-)	0.2
Negative	
:-(	-2.1
:-	-0.8

B. Sentiment Analysis

1. Sentiment Words

The system analyses the reviews to check for user sentiments associated with each comment. The system breaks the user comments into separate sentiment words [24]. A text file consists of positive and negative scores for all the positive words (good, awesome, great, fantastic, etc.) and the negative words (bad, horrible, awful, worst, terrible, etc.) respectively. Now, each sentiment word from the comment is associated with its respective score as mentioned in Table 3. The score that is associated with the word depends upon its degree of positivity or negativity. For example, the word ‘great’ has a

2. Emphasis Identification

The separated words from the comments are now checked for upper case letters, idioms or adverbs that have additional scores as they emphasize the user sentiments [25]. For example, the sentence “The food was GOOD” will have a higher score than the sentence “The food was good”, since the user has emphasized his point by using ‘GOOD’ instead of just ‘good’. Similarly, the sentence “The restaurant’s ambience was really pleasant” will have a higher score than the sentence “The restaurant’s ambience was pleasant” due the presence of the adverb ‘really’ in the first sentence thus adding an extra score to the comment.

3. Sentiment Score

The positive score and negative score for a customer review is calculated by the summation of all the positive scores and the negative scores obtained respectively and the neutral

score is calculated by taking the count of all the neutral words in a comment. The sentiment score for a comment is thus obtained by summing up all the positive, negative and neutral score. The count of exclamations and question marks is calculated and is added or subtracted from the sentiment score of a comment depending upon whether the comment is a positive comment or a negative comment respectively. For example, let us consider a positive comment and the comment consists of exclamation marks at the end. Now the number of exclamations is multiplied with a constant number and the resulting value is added to the comment's sentiment score. Similarly, in case of a negative comment, the resulting value is subtracted from the comment's sentiment score. The sentiment score is further normalized to obtain compound score in the range $[-1, 1]$. This compound score, if positive, indicates that the comment is positive and if not, a negative comment.

Table 5: Sentiment Scores for the Reviews

<i>Sentence</i>	<i>Score</i>
AWESOME WAFFLES. LOVED IT !!!!!!!	0.9124
Really GOOD. BEST ambience !!!!	0.9063

C. Rating

A rating $[1, 10]$ is given based on the compound score of every comment. The rating is updated in a way that one bad comment does not drastically affect the overall rating of an organization with previously good comments. If the rating is calculated by taking average of the recently added review's rating and the existing rating of the restaurant, there occurs a problem such that the restaurant with less number of reviews will have a higher rating than the restaurant with more number of reviews considering both have equal percentage of bad reviews. This is corrected by calculating the average by taking into account the rating of the whole set of comments which were provided from the beginning.

The rating provided depending upon this sentiment is more accurate and precise when compared to other ways of rating the restaurant. The positive comment provided indicates that the customer is satisfied and the negative comment indicates that the customer is not satisfied. Additionally neutral comments can also be provided. The sentiments derived from the customer's comments are based on four aspects: food, ambience, service and price [26].

Therefore we hypothesize the following.

- 1: The rating of a customer's comment will be higher if his/her attitude towards the taste of food in a restaurant is appreciative.
- 2: The rating of a customer's comment will be higher if he/she complements the restaurant ambience.
- 3: The rating of a customer's comment will be higher if he/she mentions a better service being provided in the restaurant in their comments.
- 4: The rating of a customer's review will be higher if they are satisfied with the pricing of the restaurant [4].

Table 6. Rating criteria

<i>Compound Score Range</i>	<i>Rating</i>
$-1.0 \leq \text{score} \leq -0.8$	1
$-0.8 < \text{score} \leq -0.6$	2
$-0.6 < \text{score} \leq -0.4$	3
$-0.4 < \text{score} \leq -0.2$	4
$-0.2 < \text{score} \leq 0$	5
$0 < \text{score} \leq 0.2$	6
$0.2 < \text{score} \leq 0.4$	7
$0.4 < \text{score} \leq 0.6$	8
$0.6 < \text{score} \leq 0.8$	9
$0.8 < \text{score} \leq 1.0$	10

4. RESULTS AND DISCUSSION

A model is trained to determine the sentiment of a restaurant review with a relatively high accuracy using the yelp reviews dataset. The system can handle thousands of reviews posted online and could sense the reviewers' emotions about any organization.

The results from this system show the relationship between the customers and the four aspects of the restaurants. The customers give reviews to the restaurant based on the four aspects, the food, the ambience, the service and the price. In addition to these four aspects, the customer's review emotions play an important role in deciding the restaurant's total rating.

The outcome of the system is the rating of a restaurant based on the sentimental analysis of the comments provided by the customers. Positive, negative and neutral score to a comment are found and sum of these scores provide the sentiment score. Compound score is the normalized sentiment score in a range of $[-1, 1]$. Every comment has a specific rating on a scale of 1 to 10 which is derived from compound score. The total rating to a restaurant is the average rating of all the comments. Single negative comment may or may not affect the rating of a restaurant with many positive comments thus providing a balance on the sentiments of all the customers who provide a comment.

Table 7. Rating Scores

<i>Sentence</i>	<i>Score</i>	<i>Rating</i>	<i>Average Rating</i>
AWESOME WAFFLES. LOVED IT !!!!!!!	0.9124	10	10
Really GOOD. BEST ambience !!!!	0.9063	10	10
AWESOME FOOD !!!!! The ice cream melts really good in your tongue. Its too good.	0.9244	10	10
Awesome place. Really good but the taste was okay.	0.7181	9	9.75
The food was really GOOD. Its a must visit amazing place !!!!	0.9316	10	9.8
bad place. but the food was really awesome. The ambience was awfully worst.	0.3071	4	8.83

5. CONCLUSION AND FUTURE WORK

Sentiment Analysis is a technique that determines whether the attitude of a person is positive, negative or neutral based on their writing [27]. The system can withstand huge volumes of customer reviews and provides better accuracy. This study is based on the data extracted from the Yelp Dataset.

This paper's contribution to the existing literature is countable. The sentiment of the customer is analyzed well in the reviews. And emphasizing sentiment scores based on the additional keywords and highlighting provides the exact replication of the customer's feelings in the rating of the restaurant.

This paper could be further studied for improvements. The theory-driven aspects on which the reviews are given could be made better by considering factors like the restaurant locality, season, the time of the year, etc. [28]. The proposed system can also be analyzed with classifiers for better results.

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